

Revised Generation Interconnection System Impact Study Report

For Queue Project Q12-0154

Jones Chapel 230 kV
March 03, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0154. The Interconnection Customer proposes a 199.1 MW wind facility interconnecting at the Jones Chapel 230 kV substation in the Indiana area. The request was received on September 02, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 113 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,950,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The POST ROCK to NEWPORT TIE 230 kV is overloaded to 106.9% of its 543 MVA emergency rating for the loss of the BEAVER to TREGO 500 kV. The project contributes 14.4% DFAX (28.7 MW) to this overload.

For the loss of BEAVER to EUFAULA 500 kV, the JOSO to ONANCOCK DIESEL GENERATION FACILITY 230 kV loads to 106.5% of its 537 MVA rating. Project DFAX: 15.6%.

POST ROCK to MYRTLE PT 230 kV: 105.9% loading (621 MVA) following the outage of the TREGO to BEAVER 500 kV; DFAX = 13.9%.

BEAVER 230/1kV XFMR: 103.2% loading (1,021 MVA) following the outage of the EUFAULA to BEAVER SUBSTATION 500 kV; DFAX = 15.1%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
POST ROCK to NEWPORT TIE 230 kV	Reconductor line	\$6,970,000	5% (\$365,000)
JOSO to ONANCOCK DIESEL GENERATION FACILITY 230 kV	Reconductor line	\$7,686,000	6% (\$423,000)
MRTLE PT to POST ROCK 230 kV	Reconductor line	\$9,312,000	5% (\$488,000)
BEAVER 230/1kV TX	Replace transformer with higher-rated unit	\$12,721,000	5% (\$674,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
	Total allocated to project		\$1,950,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.